AKASH C. BHOIR

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PROFESSIONAL SUMMARY

Manufacturing and industrial engineer with 6+ years owning the full IE scope: time study, layout design, simulation, capacity modeling, and NPI support. Comfortable on the floor and in front of leadership. Numbers follow the work.

PROFESSIONAL EXPERIENCE

Carl Zeiss X-Ray Microscopy (Zeiss XRM) — Dublin, CA
Industrial Engineer — *August 2026 – Present*

- Redesigning facility layout to align with future demand and reduce technician travel distance and non-value-added movement
- Redesigned a harness component, projecting $20K annual savings in labor and material costs
- Identified root cause of a 10-year recurring manufacturing defect on the Versa system through RCA; traced to vendor SOP gap; resolution improves long-term serviceability
- Leading pre-harnessing project and authoring SOP; projected 24% reduction in process time
- Driving technician downtime reduction initiative to improve overall line throughput
- Support manufacturing engineering team on component design in SolidWorks across Versa, Ultra, and Artesian Source product lines

Almaden — Santa Clara, CA
Industrial Engineer — *September 2024 – August 2026*

- Built greenfield assembly vertical from blank floor: MODAPTS time studies, AutoCAD layout, FlexSim simulation; achieved 48% throughput improvement and reduced travel time from 62% to 18% of cycle
- Designed production fixtures and jigs at bottleneck stations from time study data; improved repeatability and first-pass yield
- Led Kaizen, VSM, and 5S initiatives across Engineering, Quality, and Operations; drove 18% throughput improvement; managed 7 direct reports
- Built operational dashboards in Tableau and Power BI tracking cycle time, FPY, defect rates, and OTD; used Python and SQL for data consolidation and analysis
- Authored and maintained station-level SOPs, work instructions, and assembly sequences across all production stations

EVRAZ Rocky Mountain Steel — Pueblo, CO
Operational Excellence Engineer - Internship — *May 2023 – August 2023*

- Ran time studies on a pilot production line; established standard times by operation, identified cycle time losses and station imbalances, implemented process improvements achieving 8% cycle time reduction; led Kaizen blitz events targeting changeover time and WIP accumulation, developed Standard Work and visual controls to lock in gains.
- Designed a demand-driven Kanban replenishment system using SQL-based demand forecasting; reduced standing inventory 48% ($400K annual savings) while eliminating production stoppages from material shortages.
- Applied DMAIC and root cause analysis (5-Why, Ishikawa, A3) to a production bottleneck; validated results with SPC and Minitab (Cpk).

Econship Tech — Mumbai, India
Operations Executive — *March 2020 – June 2022*

- Managed production scheduling and material flow across container manufacturing operations, developed SOPs and work instructions; led a team of 4; reduced logistics costs 20% through data-driven workflow optimization.

Ashokkumar Rameshchandra & Co. — Mumbai, India
Industrial Engineer — *July 2018 – March 2020*

- Conducted time studies to establish standard times, rebalanced station workloads, and developed Standard Work documentation; implemented ABC analysis and JIT inventory practices to align stock with production demand.

EDUCATION

San Jose State University — San Jose, CA
Master of Science, Industrial and Systems Engineering — *August 2022 – May 2024*

University of Mumbai — Mumbai, India
Bachelor of Engineering, Mechanical Engineering — *July 2018*

SKILLS

Factory Analytics & Systems: Python (Pandas, NumPy, Dijkstra's Algorithm), SQL, Power BI, Tableau, Excel (Advanced, VBA, Pivot Tables), FlexSim (Discrete-Event Simulation), Minitab, SPC, ERP / MRP Systems, SAP, Multi-Source Data Consolidation, Dashboard Development & Maintenance

IE Methods & Standards: MODAPTS (Time & Motion Studies), Standard Time Development, Capacity Modeling, Takt Time Analysis, Cycle Time Analysis, Line Balancing, Labor Utilization, Hours per Unit (HPU), Work Instructions (Single Source of Truth), SOPs, Standard Work, PFMEA, Process Control Plans, Version-Controlled Documentation

Lean & Continuous Improvement: Value Stream Mapping (VSM), Kaizen / Kaizen Blitz, 5S, Gemba Walks, Standard Work, Visual Management, SMED, DMAIC, Root Cause Analysis (5-Why, Ishikawa, A3), Waste Elimination, Lean Six Sigma Green Belt

Factory Layout & Simulation: AutoCAD (2D), Factory Layout Design, Material Flow Optimization, Space Planning, Throughput & Bottleneck Modeling, Scenario Analysis, SolidWorks

NPI & Cross-Functional Execution: NPI Launch (Pilot to Full-Rate Ramp), DFM Feedback to Engineering, Live-Floor Process Transitions, Assembly Sequence Redesign, Operator Retraining, Cross-Functional Project Leadership, Stakeholder Communication, Executive Reporting

PROJECTS

Warehouse Route Optimization - Python / Dijkstra's Algorithm 2024–Present

- Building an intelligent retrieval optimization system for a 13,200-SKU, 16,600-location warehouse, implementing Dijkstra's shortest-path algorithm in Python to compute optimal picker routes and minimize travel distance; ABC velocity stratification integrated into routing priority logic; demonstrates application of advanced analytics and algorithmic optimization to manufacturing operations decision-making.

DMAIC Study of Material Unloading System - EVRAZ Rocky Mountain Steel 2023

- Led a structured DMAIC engagement to eliminate throughput bottlenecks; facilitated root cause analysis using 5-Why, Ishikawa, and A3; implemented process changes improving OEE; validated results with SPC and Minitab (Cpk); developed cost-benefit analysis to justify scope; coached operators in Standard Work to sustain gains.

All-Terrain Vehicle Competition (BAJA SAE) 2017

- Led a 12-person engineering team through end-to-end design, manufacturing, and assembly of an off-road competition vehicle; designed and fabricated welding jigs and fixtures for chassis construction; applied DFM principles; managed build schedule from raw materials through competition-ready assembly; secured 5th place in the team's inaugural season.